

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability. (BL3, BL4)*

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	5
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	5
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	5
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	5
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	5
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	5
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	5
8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	5
9	Interviewer: 'What is a serializability schedule? How do you test for conflict	BL4 – Analyze	5

	serializability using a precedence graph?		
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	5

Code of Conduct:

*I _Shaik Ayesha Tabassum(192525074)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: S.Ayesha Tabassum

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

1. Interviewer: ‘Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?’

Steps of query processing:

SQL Query → Parser & Translator → Relational algebra Expression → Query Optimizer → Best Execution Plan → Execution Engine → Result

“The query optimizer chooses the best execution plan by generating alternative plans, estimating their costs using statistics, indexes, join methods, CPU, memory and I/O requirements, and selecting the plan with the lowest estimated cost.”

2. Interviewer: What is the difference between heuristic optimization and cost-based optimization? When would you use each?

The main difference is **how the optimizer chooses the execution plan.**

Heuristic Optimization	Cost-Based Optimization
Uses predefined rules to improve the query.	Calculates the estimated cost of different plans.
Does not require detailed database statistics.	Uses statistics such as table size, number of rows, and indexes.
Faster and simpler.	More complex but usually produces better plans.
Example: Apply selection before joining tables.	Example: Choose between hash join, nested-loop join, or sort-merge join based on cost.
May not always choose the optimal plan.	Usually chooses the plan with the lowest estimated cost.

“Heuristic optimization uses predefined rules, while cost-based optimization compares the estimated costs of different execution plans. Heuristics are useful for simple and fast optimization, whereas cost-based optimization is better for complex queries and large databases.”

3. Interviewer: How would you implement database connectivity in areal project? Explain the difference between JDBC and ODBC.

JDBC vs ODBC

JDBC	ODBC
Stands for Java Database Connectivity.	Stands for Open Database Connectivity.
Mainly used by Java applications.	Can be used by applications written in different programming languages.
Uses JDBC drivers.	Uses ODBC drivers.
Designed specifically for Java.	Language-independent database connectivity standard.
Commonly used in Java web and enterprise applications.	Commonly used for connecting various applications to databases through ODBC-compatible drivers.

4. Interviewer: What are ACID properties? Give a real-world example where each property is critical.

ACID properties ensure that database transactions are processed **reliably and correctly**. ACID stands for **Atomicity, Consistency, Isolation, and Durability**.

“ACID properties make transactions reliable:

Atomicity: Transaction happens completely or not at all.

Consistency: Data always remains correct and valid.

Isolation: Transactions do not interfere with each other.

Durability: Committed data is permanently saved.

5. Interviewer: Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?

Concurrency Control Problems

- **Dirty Read:** Reading data that another transaction has changed but not yet committed.
- **Lost Update:** One transaction’s update is overwritten by another transaction.
- **Unrepeatable Read:** Reading the same data twice gives different values because another transaction changed it.

How 2PL Solves Them

2PL (Two-Phase Locking) uses **shared and exclusive locks** to control access.

- **Dirty Read** → Prevented using locks until the transaction commits.
- **Lost Update** → Exclusive locks prevent two transactions from updating the same data at the same time.
- **Unrepeatable Read** → Shared locks keep data unchanged while it is being read.

6. Interviewer: What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.

Deadlock in DBMS

- **Deadlock:** Two or more transactions wait for each other's locked resources, so none can continue.
- **Deadlock Detection:** DBMS creates a **Wait-For Graph**; if the graph contains a **cycle**, a deadlock exists.

Prevention Strategies

1. **Wait-Die:** Older transaction waits; younger transaction is rolled back.
2. **Wound-Wait:** Older transaction forces the younger transaction to roll back; otherwise it waits.

7. Interviewer: Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?

Immediate Update and **Deferred Update** are two DBMS recovery techniques that determine when changes made by a transaction are written to the database.

Immediate Update	Deferred Update
Changes can be written to the database before the transaction commits.	Changes are not written to the database until the transaction commits.
Uses UNDO and REDO during recovery.	Mainly requires REDO during recovery.
Uncommitted changes may exist on disk.	Uncommitted changes are not written to the database.
More flexible for buffer management.	Simpler recovery mechanism.
Recovery is more complex.	Recovery is comparatively easier.

8. Interviewer: Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?

Shadow Paging

Shadow Paging is a recovery technique where the DBMS keeps a **shadow copy of database pages**, so changes can be rolled back if a failure occurs.

Advantages

- No need for a **log file**.
- **Fast recovery** after a crash.
- Simple rollback.

Limitations

- Requires **extra storage** for pages.
- Page-table management can be complex.
- Less efficient for **large databases** and frequent updates.

Compared with Log-Based Recovery

Shadow Paging → simpler and faster recovery, but more storage overhead.
Log-Based Recovery → better for large and frequently updated databases, but requires maintaining logs.

9. Interviewer: What is a serializability schedule? How do you test for conflict serializability using a precedence graph?

A **schedule** is the order in which operations of multiple transactions are executed. A schedule is **serializable** if its final result is equivalent to some serial execution of those transactions.

Conflict serializability means that a schedule can be transformed into a serial schedule by swapping operations that do not conflict.

10. Interviewer: Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?

Two-Phase Locking (2PL) Variations

- **Basic 2PL:** Transaction has a **growing phase** (takes locks) and **shrinking phase** (releases locks).
- **Strict 2PL:** Holds **exclusive locks until commit/abort**, preventing dirty reads and cascading rollback.
- **Rigorous 2PL:** Holds **both shared and exclusive locks until commit/abort**.

Most Commonly Used

Strict 2PL is commonly used because it provides **strong consistency, avoids cascading rollbacks, and is practical for real-world databases.**